Food Research International 190 (2024) 114653



Contents lists available at ScienceDirect

Food Research International

journal homepage: www.elsevier.com/locate/foodres



Review



Application of the molecular dynamics simulation GROMACS in food science

Dongping Yu, Haiping Li *, Yuzi Liu, Xingqun Yang, Wei Yang, Yiran Fu, Yi-ao Zuo, Xianya Huang

Tianjin Key Laboratory of Food Biotechnology, Faculty of Biotechnology and Food Science, Tianjin University of Commerce, Tianjin 300134, China

ARTICLE INFO

Keywords:
GROMACS
Molecular dynamics simulation
Food science
Molecular interaction
Food components

ABSTRACT

Food comprises proteins, lipids, sugars and various other molecules that constitute a multicomponent biological system. It is challenging to investigate microscopic changes in food systems solely by performing conventional experiments. Molecular dynamics (MD) simulation serves as a crucial bridge in addressing this research gap. The Groningen Machine for Chemical Simulations (GROMACS) is an open-source, high-performing molecular dynamics simulation software that plays a significant role in food science research owing to its high flexibility and powerful functionality; it has been used to explore the molecular conformations and the mechanisms of interaction between food molecules at the microcosmic level and to analyze their properties and functions. This review presents the workflow of the GROMACS software and emphasizes the recent developments and achievements in its applications in food science research, thus providing important theoretical guidance and technical support for obtaining an in-depth understanding of the properties and functions of food.

1. Introduction

Food is a complex system consisting of proteins, carbohydrates, lipids and other components, all of which play a significant role in creating and maintaining food structure. Furthermore, the structures, conformations, physical and chemical properties, as well as the biological functions, of food molecules often change during food production and processing, and this has discernible effects on many of the characteristics of food, including its sensory properties, nutritional value and physiological function. Therefore, investigating the structural characteristics and intermolecular interactions of these food components can enhance food quality and function.

In recent years, the use of molecular dynamics (MD) simulation in food science has markedly increased, and this has provided valuable insights into the dynamic behavior of food molecules at the molecular level (Cen et al., 2022; Cheng et al., 2024; Meng et al., 2023; Peng et al., 2023; Tian et al., 2023). MD simulation is a promising *in silico* tool that relies mainly on the Newton's equation of motion to simulate the trajectory and state of molecular systems and samples obtained from molecular systems in various states to calculate the configuration integral of the system (Hollingsworth & Dror, 2018). Based on the results of the configuration integral, the thermodynamic quantities and other macroscopic properties of the system can be further calculated (Hollingsworth & Dror, 2018). With advances in computer technology and algorithm enhancements, MD simulation has emerged as a third method for exploring molecular structures and properties at the atomic level, following experimental methods and theoretical methods. MD simulation not only provides information that complements experimental data and assists in refining experimental designs but also serves as a powerful tool for predicting and explaining the structural and functional properties of biomolecular systems that are expensive or difficult to determine through traditional methods (Filipe & Loura, 2022). GROMACS, the most widely used molecular dynamics simulation software, possesses high computational performance and rich functional modules and can be used to simulate large-scale molecular systems and complex molecular interactions (Nian et al., 2021). In food science research, GROMACS molecular dynamics simulation can be used to visualize the structure and properties of food ingredients, such as proteins, carbohydrates and lipids, elucidate in detailed interaction mechanisms between food molecules, and observe structural changes in food molecules that occur under various experimental conditions such as changes in pH, temperature and pressure, and this can further our understanding of their roles in the flavor, texture and stability of food products. Liu, Fu, et al. (2022) studied the effects of high hydrostatic pressure and heat

* Corresponding author at: 409 Guangrong Road, Beichen District, Tianjin 300134, China.
E-mail address: hhppli@163.com (H. Li).

https://doi.org/10.1016/j.foodres.2024.114653
Received 1 February 2024; Received in revised form 14 June 2024; Accepted 15 June 2024
Available online 16 June 2024
0963-9969/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

treatment on the molecular structure of corn starch via GROMACS simulation. Gui et al. (2023) performed a 150 ns GROMACS simulation of lysozyme and cyanidin-3-O-glucoside (C3G) in which they explored the conformational changes that occur in lysozyme when C3G was added at various temperatures. Chen et al. (2023) conducted a GROMACS simulation to investigate the interaction mechanism of γ-aminobutyric acid-dioscorin complex and the effect of γ-aminobutyric acid on the structural characteristics and emulsifying properties of dioscorin. However, a systematic summary of research on the applications of the GROMACS molecular dynamics simulation software in the field of food science is currently lacking.

This paper systematically introduces the principle, workflow and advantages of the GROMACS molecular dynamics simulation software in food system simulation, and comprehensively summarizes the research progress made using GROMACS to analyze proteins, lipids, carbohydrates and bioactive small molecules in food. By summarizing recent studies and advancements, we aim to highlight the pivotal role that the GROMACS molecular dynamics simulation software plays in advancing the understand of the functions and properties of food components.

2. GROMACS molecular dynamics simulation

2.1. Overview of GROMACS

The GROMACS molecular dynamics simulation software was developed at the Department of Biophysics and Chemistry at the University of Groningen in the Netherlands in the late 1990s (Van der Spoel et al., 2005); it has since been maintained and upgraded by contributors from universities and research centers worldwide to improve its simulation capabilities. GROMACS mainly depends on the molecular dynamics, stochastic dynamics or path integral method to simulate the motion of any molecule in a system and analyze the internal interactions and the conformational changes of the system. It was originally applied to study the molecular folding and the internal interactions of the biomolecules such as lipids, nucleic acids and proteins. With advances in computer hardware and algorithms, GROMACS software is capable of simulating the systems containing millions or even billions of particles and of running multiple parallel computing methods such as message-passing interfaces (MPIs) (Abraham et al., 2015; Hess et al., 2008). This software can also employ central processing unit (CPU) and graphics processing unit (GPU) acceleration, allowing it to effectively simulate larger and more complex systems (Pall et al., 2020).

The force field is the basis for GROMACS molecular dynamics simulation, which use the obtained experimental or computational data to describe intramolecular or intermolecular interactions via potential function fitting of molecular potential energy (Mackerell, 2004). The accuracy of the GROMACS molecular dynamics simulation results depends largely on the precision of the intermolecular interaction potential. Therefore, when running simulations, it is necessary to select the appropriate molecular force field for obtaining accurate information about molecular conformational changes and molecular interactions. The classical molecular force fields consist of bonded interactions for bond stretching, angle bending and dihedral angle twisting and nonbonded interactions for electrostatic interactions and van der Waals interactions (Singh et al., 2018). The GROMACS software does not have its own force field, but it supports a variety of classical force fields (Van der Spoel et al., 2005). Table 1 lists the force fields that are widely used in food research, including AMBER (Assisted Model Building with Energy Refinement), CHARMM (Chemistry at Harvard Macromolecular Mechanics), GROMOS (Groningen Molecular Simulation) and OPLS (Optimized Potentials for Liquid Simulations) (González, 2011).

2.2. Procedure of simulation

The procedure followed in the GROMACS simulation can be generally divided into four steps: preparation, setup, simulation and analysis, as depicted in Fig. 1. Although the simulation process is completed through a set of commands and files, each step must be meticulously designed and rigorously executed to ensure that the simulation results accurately reflect the experimental data.

Table 1
Application of force fields for food research.

Force field	Applicable molecular classes	Where to get
OPLS-AA	Proteins, carbohydrates, lipids, organic small molecules	https://zarbi.chem.yale.edu/oplsaam.html
CHARMM	Proteins, carbohydrates, lipids	https://mackerell.umaryland.edu/charmm_ff.shtml#gromacs
GROMOS	Protein, carbohydrates, lipids (membrane), small molecules	https://vienna-ptm.univie.ac.at/?page_id=100
AMBER	Protein, carbohydrates, lipids	https://ambermd.org/AmberModels.php

2.2.1. Preparation

Before performing a simulation, a molecular structure file (pdb file or gro file) and a molecular topology file (top file) must be obtained. The pdb file contains information such as the initial coordinates, serial numbers and residue names of each atom in the molecule. The gro file, a file type that is unique to GROMACS, also contains coordinate information on the molecules. The most significant distinction between the gro file and the pdb file is that the gro file can also store speed information. The top file provides detailed information on molecular structure, including atom types, charges, and the presence of bonds, angles and dihedrals (Smith et al., 2022). The initial protein structure files are usually retrieved from the Protein Data Bank, and the initial structures of common small-molecule compounds can be obtained from the PubChem and ChemSpider websites (Chen et al., 2019). Molecular structures that cannot be obtained from these databases can be built using online websites such as CHARMM-GUI and GLYCAM or software packages such as ChemOffice and Materials Studio. However, obtaining accurate molecular structures and validating interaction parameters can be challenging. This process often requires careful inspection and extensive validation to ensure the accuracy and reliability of the structures and parameters. The next step is to choose the appropriate force field, such as AMBER, CHARMM, GROMOS or OPLS. The final step in the preparation for a simulation is to convert the “topol.top” topology file and the “x.gro” coordinate file using the command “gmx pdb2gmx -f input.pdb -o output.gro -p topol.top”, but this command is confined to molecules and residues that are defined in the selected force field files (Hu et al., 2023). For molecules that are not defined in standard force fields, researchers can supplement the force field files by defining new atom types, bonds, angles and other specific parameters of the molecule. One can also utilize external parameterization tools such as Sobtop, CGenFF, PRODRG or ATB to generate GROMACS-compatible topology files for specific types of systems. In some cases, creating or editing topology files may be necessary, and this may require significant expertise in molecular modeling.

2.2.2. Setup

The setup includes defining the simulation box and adding the solvent water and ions. Most common simulations are conducted under periodic boundary conditions (PBC) to avoid boundary effects caused by the finite size of the simulation box (Dondapati, 2019). PBC helps create an infinite system by allowing particles that exit one side of the box to reenter from the opposite side (González, 2011). The simulation box can be created by the command “gmx editconf -f input.gro -o output.gro -bt box-type -box x y z”. Common box types include cubic, rhombic dodecahedron and truncated octahedron. It is important to note that most of the systems studied in food science are in aqueous environments. Therefore, to simulate the experimental conditions more accurately, it is

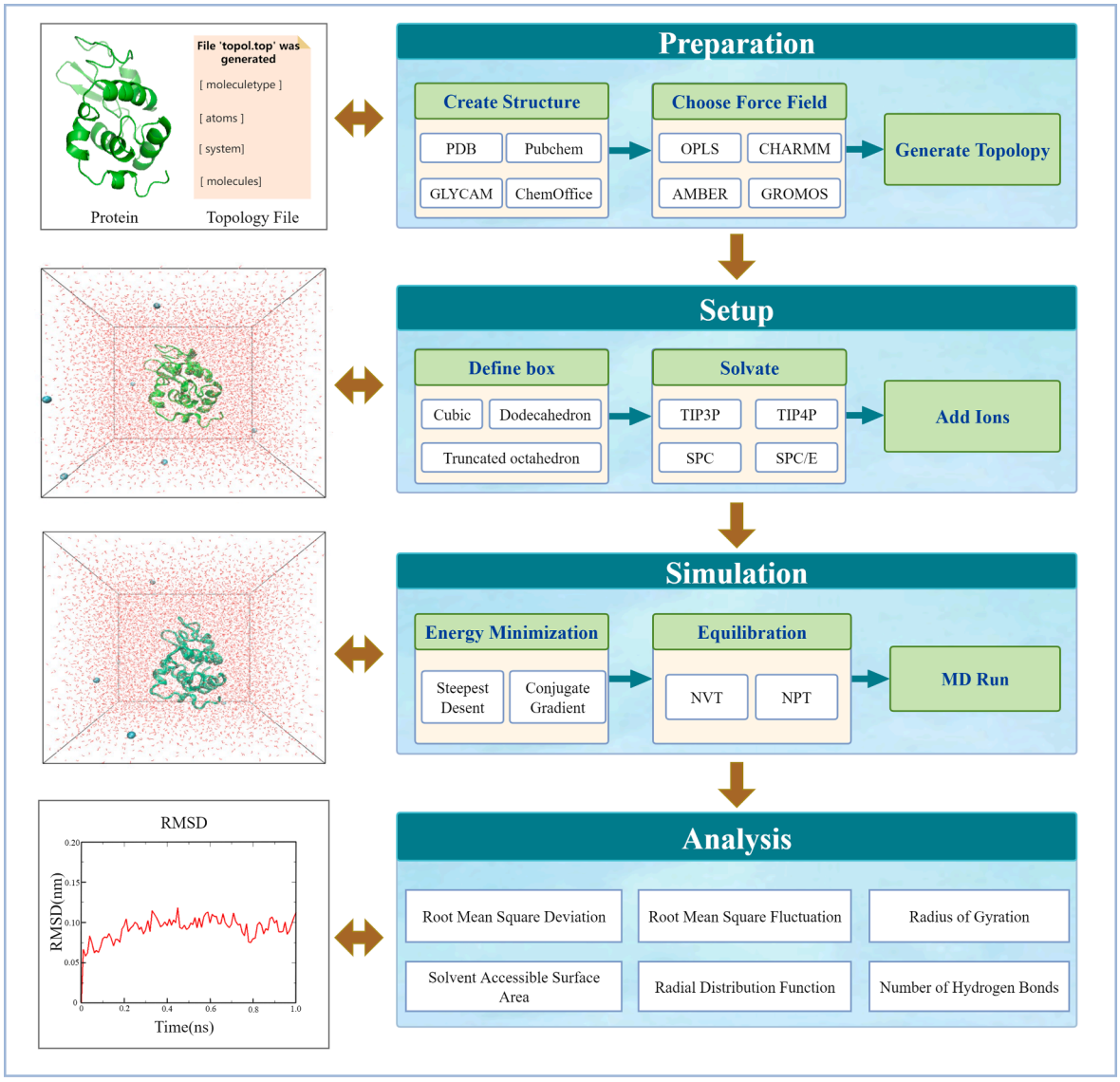


Fig. 1. The workflow of GROMACS molecular dynamics simulation in food science.

necessary to add water (Smith et al., 2022). The command "gmx solvate -cp input.gro -cs water.gro -p topol.top -o output.gro" can be used to fill the system with solvent and select the water type. The optional water models are TIP3P, TIP4P, TIP5P, SPC and SPC/E. In order to make the system electrically neutral, a certain number of counter ions should be added. Next, a molecular dynamics parameter file (mdp file) is required as the input file. The mdp file contains all information about the molecular dynamics simulation itself, e.g., time step, number of steps, boundary conditions, temperature, pressure and other parameters. The command "gmx grompp -f ions. mdp -c input.gro -p topol. top -o ions. tpr" combines coordinate and topology information with the parameters set in the mdp file to generate the corresponding binary input file (tpr file); the command "gmx genion -s ions.tpr -o output.gro -p topol.top -pname positive-ion -nname negative-ion -neutral" can then be applied to specify the type and number of ions.

2.2.3. *Simulation*

Energy minimization is a key step in the GROMACS simulation, aiming to minimize the abnormal phenomena within the constructed system and move the system in the direction toward the nearest local energy minimum (Smith et al., 2022). The methods commonly used for energy minimization are the steepest descent method and the conjugate gradient method. Similarly, it is necessary to set the MD simulation parameters in a mdp file for simulation process. The command "gmx grompp - f em.mdp - c input.gro - p topol. top - o em. tpr" generates the tpr file, and the command "gmx mdrun -deffnm em" is then used to minimize energy. After energy minimization, pre-equilibration should be conducted to adjust the molecular structure and enhance the stability of the system (Feng et al., 2015). This phase consists of different ensembles, including the micro-canonical (NVE) ensemble, which maintains a constant number of atoms (N), constant volume (V) and constant total energy (E); the canonical (NVT) ensemble, which maintains a constant number of atoms (N), constant volume (V) and constant temperature (T); and the isothermal-isobaric (NPT) ensemble, which maintains a constant number of atoms (N), constant pressure (P) and constant temperature (T). In food science, the NVT and NPT ensembles are commonly used for simulations and analyses. When the system reaches equilibrium, the command "gmx mdrun -deffnm md" can be ran the dynamics simulation under different conditions (Smith et al., 2022). The simulation running time ultimately depends on the complexity of the target system and the processing performance of the computer used. Reproducibility of simulation results is crucial in ensuring the reliability of scientific research. It is necessary for researchers to perform multiple repeated simulations to ensure that the results are not obtained by

chance. To achieve reproducibility, it is of the utmost importance to exercise strict control over the initial conditions and input files, to utilize the same random seed for the generation of initial velocities, to make sure consistency in the GROMACS software version and to maintain identical hardware configurations and parallel computing settings. Additionally, verification of the repeatability of the simulation results using different types of MD software is encouraged.

2.2.4. Analysis

The GROMACS molecular dynamics simulation provides a number of analytical tools that can be used to calculate various properties of the simulation outcomes, such as root mean square deviation (RMSD), root mean square fluctuation (RMSF), radius of gyration (Rg), solvent accessible surface area (SASA), and number of hydrogen bonds (NHB) (Smith et al., 2022). Detailed descriptions of these parameters and the commands used to obtain them are presented in Table 2. At the same time, using visualization software such as VMD and PyMOL, trajectory data are displayed as graphics or animation that can be used to intuitively observe the state of motion of the system and analyze various macroscopic and microscopic properties of the system (Chen et al., 2019).

2.3. Advantages of GROMACS

The use of GROMACS for MD simulations in food science research has several notable advantages. GROMACS is a highly optimized software that exploits the parallel processing capabilities of modern computers to achieve large-scale molecular simulation. Its running efficiency is superior to that of several mainstream similar software programs, includng AMBER, CHARMM and LAMMPS, and its simulation speed is also 3 ~ –10 times faster than that of similar software programs (Hu et al., 2023). This software makes it possible to perform fast and efficient calculations when investigating food systems that contain large numbers of molecules. It also provides rich simulation functions that accurately describe the structure and properties of food components such as proteins, lipids and carbohydrates (Chen et al., 2019). Moreover, the GROMACS software supports the use of various classical force fields and simulation techniques and thereby allows researchers to obtain accurate and reliable simulation results (Van der Spoel et al., 2005). In addition, the GROMACS software is an open-source software with broad compatibility, characteristics that are preferred by many users. The software can be integrated with a range of molecular simulation tools and visualization software to aid researchers in the processing, analysis and visualization of data, thus providing more choices and flexibility for food researchers.

GROMACS software also has a wide range of applications in food safety and the food industry (Condict et al., 2021; Wang, Zhu, et al., 2022; Yu et al., 2023; Zhang et al., 2018; Zhu et al., 2022). Through simulation using GROMACS, researchers can obtain an in-depth understanding of the functions and properties of the molecules in food at the micro level, which is helpful for addressing many problems in food science and food technology. Therefore, GROMACS serves as an efficient, accurate and compatible molecular simulation software, providing a reference and technology for food research and advancing the field of food science.

Table 2
Analysis parameters of GROMACS commonly used in food science.

Analysis parameter	Characterization	Command
Root mean square deviation, RMSD	Measured whether the simulation system reaches the equilibrium; when the RMSD value fluctuates within a small range, the molecular structure can be considered balanced.	gmx rmsd
Root mean square fluctuation, RMSF	Measured the stability of the molecular conformation; when the RMSF value of the system is smaller, it indicates the greater convergence and the higher stability.	gmx rmsf
Radius of gyration, Rg	Characterized the tightness of biomolecules; the lower RG value indicates the tighter molecular structure.	gmx gyrate
Solvent accessible surface area, SASA	Described the total surface area in which the molecules can contact the solvent.	gmx sasa
Radial distribution function, RDF	Described the distribution of the distances between the molecules or the atoms, reflecting the spatial arrangement and the interaction of the molecules or the atoms.	gmx rdf
Number of hydrogen bonds, NHB	Computed and analyzed the number of hydrogen bonds. This is crucial for understanding the intermolecular interactions in biomolecules.	gmx hbond

3. Application of GROMACS in the food science

The primary nutritional components of food comprise proteins, lipids and carbohydrates, all of which possess unique structures and nutritional functions. Structural alterations in food molecules that occur during food production and processing significantly affect the color, taste and texture of the final product. Consequently, it is necessary for scientists to delve into the dynamic behaviors of food components. GROMACS serves as a powerful tool for studying molecular architectures and functional characteristics in food science at the atomic scale, thereby enhancing our understanding of the properties and functions of food molecules and providing valuable insights in improving food quality and developing new food products. Table 3 lists the recent applications of the GROMACS molecular dynamics simulation software in food research, which summarizes four aspects including food molecular system, molecular type, force field/water model and simulation function.

3.1. Protein

Proteins are one of the major components of food, and their functional properties are intrinsically related to their structure. Protein conformation is often complex and can undergo dynamic changes during food processing, making it very difficult to investigate using traditional methods (Singh et al., 2018). However, through GROMACS molecular dynamics simulation, various essential processes such as folding, denaturation and conformational transitions of protein can be observed, which enables researchers to gain crucial insights into the structural stability and dynamic behavior of proteins (Euston, 2013; Stănciuc, Creţu, et al., 2018; Zhou et al., 2022). Nnyigide and Hyun (2020) studied the heat-induced denaturation of bovine serum albumin and the protective effect of sodium dodecyl sulfate on denaturation using GROMACS simulation. Han et al. (2022) conducted a simulation of the myosin-sesamol complex to investigate the impact of sesamol on myosin structure. The authors found that hydrogen bonding and hydrophobic interactions between the two substances caused the myosin protein to become more compact, thereby significantly enhancing its stability. Abdollahi et al. (2023) discovered that the binding site of the ferulic acid to the β-lactoglobulin was located between the FG-loop and the 310 helical turn, which caused a change in the conformational structure of the protein in the complex, ultimately improved the stability of the β-lactoglobulin structure.

During food processing and storage, the functional properties of proteins are highly susceptible to the effects of heat, chemicals, radiation or mechanical treatments (Corredig et al., 2020). However, the traditional techniques such as circular dichroism, nuclear magnetic resonance and X-ray diffraction often fail to reveal the conformational changes that occur during these processes, rendering them inadequate to provide a comprehensive explanation of how dynamic changes in protein structure affect the functional properties of the proteins (Chen et al., 2019). By utilizing the GROMACS simulation, dynamic alterations in proteins that occur under various processing conditions can visualized,

Table 3
The application of the GROMACS in food research.

System	Target Molecule	Force field /Water model	Function	References
Protein	Bovine serum albumin	–	It is used to observe the processes of the protein denaturation and conformational change, and uncover the structural stability and the dynamics behavior of the proteins.	Nnyigide & Hyun, 2020
	Myosin	CHARMM36/TIP3P		Han et al., 2022
	β-Lactoglobulin	CHARMM27/TIP3P		Abdollahi et al., 2023
	Rhizopus chinensis lipase	GROMOS43A1/SPC	It can visualize the dynamic alterations of the proteins under various food processing conditions.	Chen et al., 2017
	Gliadin	GROMOS43A1/–		Stănciuc, Banu, et al., 2018
	ι-actoferrin	GROMOS54A7/SPC		Darmawan et al., 2020
	Yam soluble protein	OPLS-AA/TIP3P		Nie et al., 2023
	Trypsin	CHARMM36/–		Li et al., 2022
	Bovine serum albumin	AMBER99/TIP3P		Yang et al., 2023
	α-Amylase	CHARMM36/–	It serves as an effective virtual screening tool to elucidate the intricate interaction between the enzymes and the substrates.	Ye et al., 2022
	α-Glucosidase	AMBER99/SPC		Wang, Fan, et al., 2023
	α-Glucosidase inhibitory peptides	AMBER99/TIP3P		Wang, Deng, et al., 2023
	Endo-β-1,4-xylanase	GROMOS54A7/SPC		Wu et al., 2023
	β-Lactoglobulin	CHARMM36/SPC	It plays a key role in studying the interaction between the protein–protein and/or the protein–ligand.	Di & Jia, 2023
	Grape seed 11S globulin	GROMOS/SPC		Chamizo-González et al., 2023
	Pumpkin seed protein	AMBER99/TIP3P		Liang et al., 2023
Carbohydrate	Methylcellulose	GROMOS56A$_{CARBO}$/ GROMOS45A4/SPC	It can describe the dynamics behavior of the hydrates and explore the structural stability of the carbohydrate molecules in solution.	Huang et al., 2014
	Glucose	GROMOS56A$_{CARBO}$/SPC/E		Batista et al., 2015
	Single-stranded curdlan	CHARMM36/TIP3P		Feng et al., 2020
	κ-Carrageenan	GLYCAM/SPC/E		Westberry et al., 2022
	Gellan gum	CHARMM36/TIP3P		Yan et al., 2023
	V-type linear dextrin	CHARMM36/–	It is critical for elucidating the formation and the stabilization mechanism of the inclusion complexes between the carbohydrates and the guest molecules.	Liu, Junejo, et al., 2022
	β-Cyclodextrin	CHARMM36/TIP3P		Deng et al., 2022
	Cyclodextrin	GROMOS96/SPC		Kumar et al., 2023
	Amylose	GROMOS53A6/SPC	It is conducive to a deeper study of the microscopic interaction mechanism between the carbohydrate and other food components.	Cheng et al., 2018
	Amylose	OPLSAA/TIP4P		Xu et al., 2022
	Carboxymethyl β-glucan	AMBER99/TIP3P		Wang, Yu, et al., 2022
Lipid	1-Naphthol-loaded nanoparticle	–	It can be used to observe the formation process of the lipid nanoparticle systems.	Bhopatkar et al., 2015
	Soybean oil-based nanoemulsion	CHARMM/TIP3P		Moghaddasi et al., 2018
	Solid lipid nanoparticle	OPLS-AA/SPC/E		Guo et al., 2022
	Pickering emulsion gels	–/SPC	It is capable of describing the microscopic behavior and the mechanical properties of the emulsion gels.	Geng et al., 2021
	Gelator dimer	–		Jiang et al., 2020
	POPC	GROMOS53A6/SPC	It can characterize and investigate the dynamics and the detailed mechanism of the interaction of the biomolecules and the biofilms	Neto & Cordeiro, 2016
	POPE/POPG/POPC /DOPC/DPPE	CHARMM36/TIP3P		Lyu et al., 2018
	DPPC	GROMOS43A2/SPC		Alghamdi et al., 2019
	POPC/POPG	GROMOS53A6/SPC		Nie et al., 2021
Small molecules	Naringenin	GROMOS96/SPC	It provides valuable guidance for understanding the embedding mechanisms of the bioactive compounds and for screening the optimal protein carriers of these substances.	Gholami & Bordbar, 2014
	Carotenoid	GROMOS43A1/–		Aprodu et al., 2017
	Lycopene	CHARMM36/SPC		Wang et al., 2021
	Cyanidin-3-O-glucoside	GROMOS43A1/SPC		Pan et al., 2021
	Sunset yellow	GROMOS43A1/TIP3P	It can visualize the three-dimensional conformations and the binding mechanisms between the food additives and the biological molecules.	Mohseni-Shahri et al., 2018
	Tartrazine	AMBER99/TIP4P		Jafari-Arvari et al., 2021
	Quinoline yellow	AMBER99/TIP4P		Hashemi-Shahraki et al., 2021
	Ponceau 4R	CHARMM36/SPC/E		Sadat Mostafavi et al., 2022

Abbreviations: POPC, 1-palmitoyl-2-oleoyl-1-palmitoyl-*sn*-glycero-3-phosphocholine; POPG, 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphoglycerol; POPE: 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphoethanolamine; DOPC: 1, 2-dioleoyl-*sn*-glycero-3-phosphocholine; DPPE: 1, 2-dipalmitoyl-*sn*-glycero-3-phosphoethanolamine; DPPC: diphosphatidyl-choline; CHARMM: chemistry at harvard macromolecular mechanics; AMBER: assisted model building with energy refinement; GROMOS: groningen molecular simulation; OPLS: optimized potentials for liquid simulations; TIP3P: transferable intermolecular potential 3 points; TIP4P: transferable intermolecular potential 4 points; SPC: simple point charge; SPC/E: extended simple point charge.

which can guide researchers in determining the most favorable conditions for the preservation of proteins during food processing and storage. Chen et al. (2017) employed MD simulation to investigate the impact of high hydrostatic pressure on the conformation of the *Rhizopus chinensis* lipase (RCL). Utilizing the GROMACS software, the RCL enzyme was subjected to the different pressures (0.1, 200, 400 and 600 MPa) and then the enzyme's conformational stability, lid movement and active site change were analyzed. The study revealed that a stable and partially open lid conformation was maintained at pressures below 400 MPa, thus enhancing the enzymatic activity. In contrast, the lid exhibited a more closed conformation at 600 MPa, leading to weakened interactions within the active site and reduced the enzymatic activity. Stănciuc, Banu et al. (2018) found in their MD simulation that thermal treatment significantly altered the structural properties and antigenic properties of gluten proteins. Darmawan et al. (2020) simulated the effect of spray drying temperature on the complex of lactoferrin and whey protein. They found that addition of whey protein increased the stability of lactoferrin under high-temperature conditions. The stability of lactoferrin under freeze-drying conditions was also confirmed, which verified the possibility of the protein components encapsulating the lactoferrin (Darmawan et al., 2021). The effects on the structural changes and the emulsifying properties of yam soluble protein were also studied through GROMACS molecular dynamics simulations under freeze-drying and spray-drying temperatures (Nie et al., 2023). Li et al. (2022) explored the effect of pulsed electric field (PEF) treatment on the spatial conformation and activation mechanism of trypsin. As shown in Fig. 2, the results for root mean square deviation (RMSD), solvent accessible surface area (SASA), radius of gyration (Rg) and number of hydrogen bonds indicated that after PEF treatment molecular structure of the trypsin was more stable and the binding affinity of the trypsin-substrate complex was stronger. Yang et al. (2023) investigated the mechanism of binding of the anthocyanin-protein complex and its pH dependence using bovine serum albumin and cyanidin-3-O-glucoside as models. The results showed that the binding affinity between the C3G and the BSA was significantly influenced by pH value, and that electrostatic interactions played a significant role in the binding, particularly at higher pH.

As natural biocatalyst, enzymes have been widely applied in food processing. They catalyze a wide range of chemical reactions at various stages of food production and processing, which contributes to several beneficial outcomes in the food industry, including extension of the shelf life of food products, improvements in the taste and quality of food, and optimization of the efficiency of food production (Huang et al., 2023). The GROMACS simulation provides an effective virtual screening tool for elucidating the intricate interactions between enzymes and substrates, and can predict critical aspects such as the location of the active site, the catalytic mechanism and the substrate specificity of the enzyme. Ye et al. (2022) simulated the inhibition mechanism of α-amylase by L-arginine. The authors found that L-arginine was present only on the surface of the enzyme and that it did not enter the hydrophobic cavity of the enzyme during the simulation. Thereby, it was demonstrated that L-arginine could reduce the activity of the enzyme by inducing α-amylase unfolding. Wang, Fan, et al. (2023) explored the inhibitory effect of polydatin and resveratrol on the α-glucosidase. Molecular dynamics simulations revealed that the α-glucosidase-polydatin and α-glucosidase-resveratrol complexes were stabilized by the hydrogen bonding. Moreover, polydatin and resveratrol entered the active site of α-glucosidase, preventing the binding of α-glucosidase to its substrate and thereby inhibiting the enzyme's activity. Wang, Deng, et al. (2023) screened the α-glucosidase inhibitory peptides present in *Ginkgo biloba* seed cake by GROMACS simulation and verified that they had good

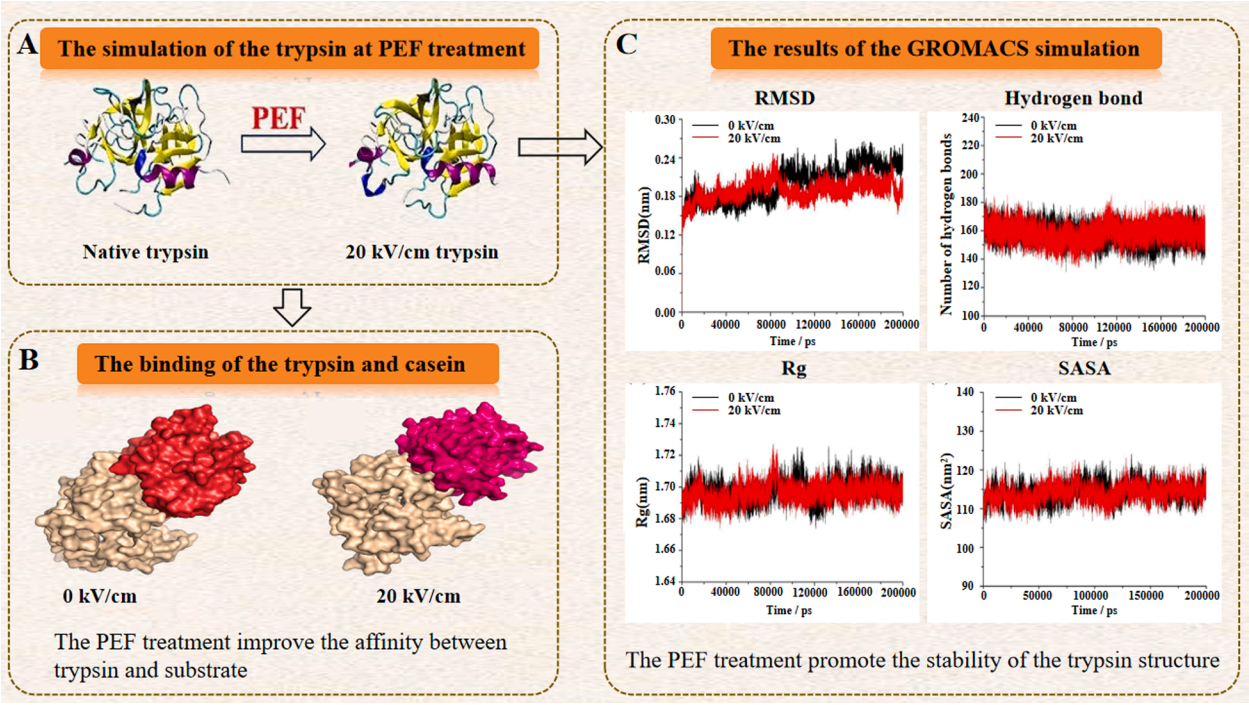


Fig. 2. The effect of PEF treatment on the structure of trypsin (Li et al., 2022). (A) The conformational change of trypsin after PEF treatment; (B) The binding conformation of trypsin and substrate at 0 kV/cm and 20 kV/cm; (C) The value of RMSD, Rg, SASA and number of hydrogen bonds of trypsin at 0 kV/cm and 20 kV/cm over time.

α-glucosidase inhibitory capacity. Wu et al. (2023) predicted and analyzed the flexible region of the *endo*-β-1,4-xylanase using GROMACS software. Their findings revealed that the N-terminal flexible ring region plays a crucial role in augmenting the catalytic activity of GH11 xylanase. Furthermore, this region was identified as a potential target for enhancement of the enzyme's catalytic performance. Through the integration of multiple effective sites, they successfully engineered a combinatorial mutant enzyme. This mutant exhibited not only improved catalytic activity but also enhanced thermal stability.

The atomic interactions between proteins and those between proteins and ligands have been widely studied in recent years via GROMACS molecular dynamics simulations (Abdollahi et al., 2022; Shao et al., 2018). Di and Jia (2023) reported the interaction mechanism between the β-LG and the flavor components in the goat milk through the molecular dynamics simulation; the results showed that the protein conformation changed after interactions between these components. To assess the stability of the molecular interaction between grape seed 11S globulin present in winemaking by-product and malvidin-3-O-glucoside, a GROMACS simulation was performed (Chamizo-González et al., 2023). They found that a stable complex was formed at the most hydrophobic site of the protein for which van der Waals forces and hydrogen bonds were the main driving forces. Liang et al. (2023) utilized GROMACS software to determine the stability of the binding of pumpkin seed protein to apigenin and to understand the dynamic interactions involved.

GROMACS molecular dynamics simulation is not only essential for observing conformational changes in protein molecules and the stability of protein–ligand binding but also contributes to prediction of enzyme catalytic sites and substrate screening, thus promoting a better comprehension of protein functional properties. Furthermore, the use of GROMACS molecular dynamics simulation to explore the dynamic behaviors of a protein under different experimental conditions provides a directive reference for food processing that helps identify optimal processing conditions and reduces research costs.

3.2. Carbohydrate

Carbohydrates are hydrophilic; the majority of the functional groups in carbohydrate molecules form hydrogen bonds with water molecules through hydroxyl groups (Feng et al., 2015). Due to this interaction, carbohydrates have a strong affinity for the solvent water. Consequently, carbohydrates can influence the structure of the surrounding water, and conversely, water can also impact the conformation of carbohydrate molecules (Feng et al., 2015). This characteristic has a significant effect on properties such as viscosity, stability and gelatinization (Chen et al., 2019). Although many scientific investigations have explored the hydration of carbohydrates, these investigations have been unable to provide detailed information on hydration at the atomic level. GROMACS molecular dynamics simulation can describe the behavior of hydrates, and explore the structural stability of carbohydrate molecules in solution. Huang et al. (2014) analyzed the solvation and gelation behavior of methylcellulose (MC) in aqueous and acetone solvents using MD simulations with GROMOS 56_{ACARBO} and GROMOS 45A4 force fields. The following conclusions were reported: the unmethylated O6 position on the cellulose ring formed the most hydrogen bonds, and methylation at this position significantly reduced the hydrogen bonding. Dimethylated and trimethylated MCs formed aggregate structures at lower temperatures and precipitate-like structures at higher temperatures in water, whereas they dispersed randomly in acetone. The gelation behavior of the methylcellulose was primarily driven by hydrophobic interactions. Batista et al. (2015) conducted GROMACS simulations for the six mixtures between glucose and water at different concentrations to evaluate the accuracy of the GROMOS $56A_{CARBO}$ force field for modeling aqueous glucose solutions. Feng et al., 2020 investigated the conformational properties of single-stranded curdlan with different chain lengths in aqueous solution by performing simulations using the CHARMM36 force field and GROMACS software. The study concluded that the helical conformation of curdlan in water was influenced by multiple factors, including its glycosidic linkages, water solvation and hydrogen bonding. Westberry et al. (2022) used MD simulations combined with X-ray scattering to explore the secondary structure of κ-carrageenan in solution. Their study challenged the long-held view that κ-carrageenan exists as a "random coil" in solution. Instead, κ-carrageenan appeared to retain many helical motifs in solution and to maintain a significant degree of secondary structure that was similar to the structure it exhibited in the solid state. Yan et al. (2023) investigated the conformation and hydration properties of low-acetyl gellan gum under microwave irradiation utilizing MD simulation. The simulation demonstrated that microwave irradiation did not significantly alter the structural conformation of the gellan molecule but that it did affect its interaction with water molecules. Under microwave irradiation, the hydration shell of the gellan molecule became more compact, and there was an increase in the lifetime and the average number of hydrogen bonds.

Carbohydrates usually form inclusion complexes with guest molecules, effectively increasing the water solubility, structural stability and biological activity of these molecules (Premjit et al., 2022). This process can also mask undesirable flavors, thus improving the sensory quality of food. Understanding the interaction between carbohydrates and guest molecules will contribute to the use of inclusion complexes in food (Feng et al., 2015). GROMACS simulation is critical for elucidating the mechanism of formation and stabilization of inclusion complexes between carbohydrates and guest molecules. Liu, Junejo, et al. (2022) investigated the effect of crystallization conditions on the formation of V-type linear dextrin (LD) and its ability to encapsulate ethylene gas. Molecular dynamics simulation was employed to explore how different solvents (water in the MI system and ethanol in the MII system) influenced the conformational behavior of the LD molecules. A random coil structure was formed in the MI system, and a single helix structure was formed in the MII system, indicating the potential of V-type linear dextrin as a carrier for ethylene gas. Deng et al. (2022) conducted MD simulations study of the formation and stabilization of inclusion complexes (ICs), focusing on the interaction between β-cyclodextrin (β-CD) and five different C10 aromatic molecules. As depicted in Fig. 3, the results of the MD simulation, including RMSD, Rg, SASA and number of hydrogen bonds, provided a comprehensive understanding at the molecular level of the mechanisms through which these ICs are formed and stabilized. Kumar et al. (2023) evaluated the stability of the inclusion between chrysin (ChR) and various forms of cyclodextrin (α-CD, β-CD and γ-CD) with the goal of identifying the form of CD that is most suitable for improving the hydrosolubility of the poorly soluble ChR. Their study confirmed that γ-CD, which has the largest central cavity, was the most suitable for enhancing the solubility of ChR.

Carbohydrates play an essential role in food processing by interacting with other food components, thereby contributing to enriching the flavor and nutritional characteristics of food products. Conventional experimental techniques can unveil the ultimate structure of the complex but cannot depict the dynamic changes that occur during the binding process (Chen et al., 2019). Consequently, GROMACS simulation is increasingly employed to study the mechanism of microscopic interaction between carbohydrates and other food components. Cheng et al. (2018) investigated the stability and conformation of complexes formed during the interaction between amylose and linoleic acid at 300 K by MD simulation. The results showed that a stable complex was formed in which linoleic acid was located in the cavity of the amylose helix. They also observed that the intra-molecular hydrogen bonds within the amylose helix contributed to the stability of the complex. Xu et al. (2022) studied the physicochemical and morphological properties of corn starch when combined with three different charged amino acids (AAs) (glycine (Gly), lysine (Lys) and glutamic acid (Glu)). The RMSD analysis indicated the conformational stability of the starch molecular chains was improved with addition of AAs. Analysis of the dihedral

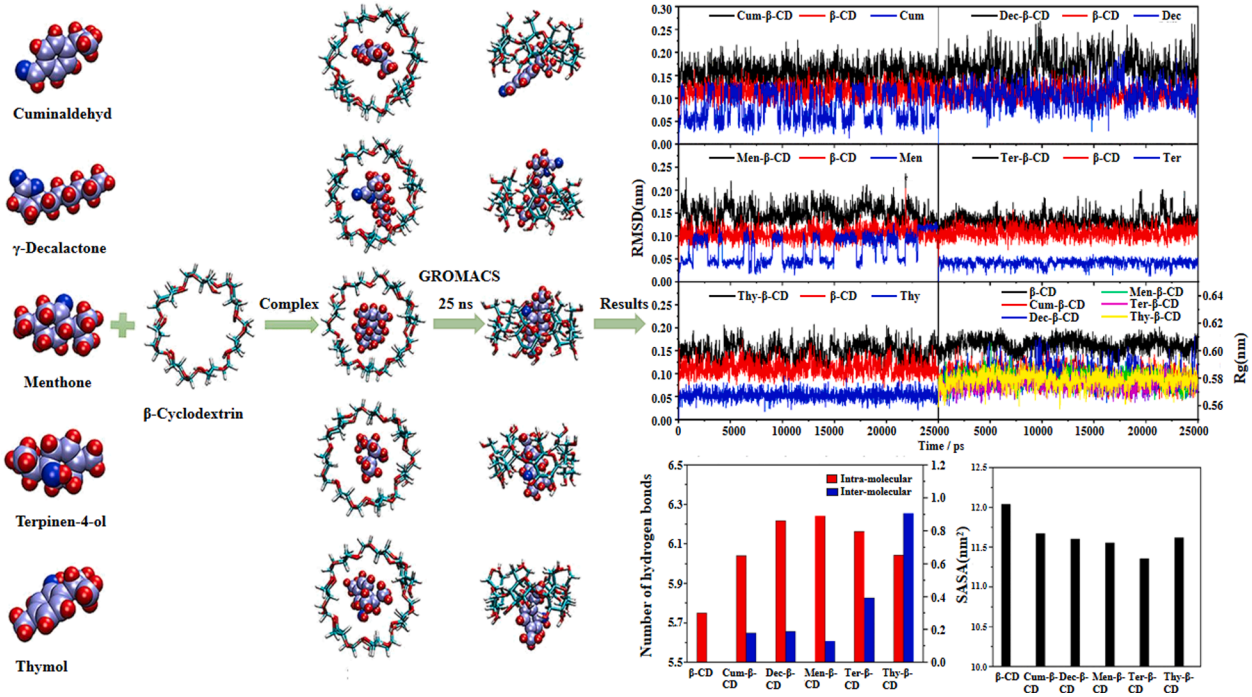


Fig. 3. The conformational changes and results of the interaction between β-cyclodextrin and different aroma molecules after GROMACS molecular dynamics simulation (Deng et al., 2022).

angles revealed that the spatial configuration of the starch molecules became more stable when they were combined with Glu and Lys; this was attributed to the increased formation of intramolecular hydrogen bonds by these AAs, resulting in reduced dihedral angles. As a result, the physical and chemical properties of the starch were altered. Specifically, the swelling capacity of the starch decreased, and its gelatinization temperature increased. Wang, Yu, et al. (2022) explored the mechanism of binding between anionic carboxymethyl *Poria cocos* polysaccharide (CMP_N) and epicatechin gallate (ECG). The molecular dynamics simulations suggested that ECG combined with CMP_N through non-covalent interactions that included hydrogen bonding, hydrophobic interactions and electrostatic interactions, which led to changes in the apparent potential, hydrophobicity and particle size of the complex and ultimately had a synergistic antibacterial effect.

GROMACS simulation has proven to be an effective technology for gaining in-depth insight into the hydration, solvation and gelation behavior of carbohydrates and for exploring the formation and stability of inclusion complexes between carbohydrates and guest molecules, as well as for elucidating the mechanisms through which carbohydrates interact with other food ingredients. The results of the simulations provide valuable guidance to researchers on how to improve the texture, stability and taste of food products.

3.3. Lipid

The application of GROMACS molecular dynamics simulation is of great importance in probing the functions and properties of food lipids at microsecond time scales. The encapsulation and delivery of active compounds in food through nanoemulsions and solid lipid nanoparticles can improve the texture, stability and nutrition of food products, and this has become a popular research topic in food industry in recent years (Da Silva Santos et al., 2019; Dasgupta et al., 2019; K. & Kumar, 2022). Few investigations of the underlying formation mechanism of these nanostructures have been conducted due to experimental limitations. However, GROMACS molecular dynamics simulation can be used to observe the process through which nanoemulsions and solid lipid nanoparticles are formed. Bhopatkar et al. (2015) performed a GROMACS simulation in which they attempted to understand the self-assembly process and the incorporation mechanism of the 1-naphthol into soluble nanoparticles prepared from starch, protein and fatty acids. The MD simulation confirmed that the self-assembly process was favorable and that the 1-naphthol molecule was located at the helix of the amylose lumen. Moghaddasi et al. (2018) employed molecular dynamics simulations to investigate the formation of nanoemulsions in the presence and absence of curcumin. This study provided detailed insight into how curcumin nanoemulsions self-assemble at the atomic level. The simulations revealed that the curcumin nanoemulsion particles have spherical or nearly spherical shapes. The addition of curcumin decreased the amount of time required to reach equilibrium and resulted in a more symmetrical and compact system. Guo et al. (2022) explored the ability of solid lipid nanoparticles to deliver phytosterols. The GROMACS simulation results suggested that the affinity between phytosterols and glycerides decreased in the following order: glycerol distearate > glycerol tristearate > glycerol monostearate. This finding was important in explaining the release and storage capacities of the three types of nanoparticles.

The microscopic behavior and mechanical properties of emulsion gels can be characterized using GROMACS molecular dynamics simulations. Geng et al. (2021) conducted a GROMACS simulation to elucidate the microscopic behavior of dihydromyricetin in an oil–water system. Molecular dynamics simulations confirmed that dihydromyricetin, which was initially randomly distributed in water, eventually aggregated and adsorbed to the oil–water interface. This contributed to the formation of a stable system, indicating that dihydromyricetin has the potential to stabilize the Pickering emulsion gels. Jiang et al. (2020) explored the self-assembly behavior of the gelators of stearic acid (SA), 12-hydroxy stearic acid (HSA) and 2-hydroxyethyl stearate (HES). The HSA dimer exhibited the lowest values of $E_{Interaction}$ and Rg indicating that the HSA dimer was the most stable and that the oleogels formed with HSA possessed the most favorable mechanical properties.

Biofilms formed by pathogenic microorganisms and decaying microorganisms are a primary source of food contamination, and they lead to significant food quality and safety issues and pose a threat to human health (Alvarez-Ordonez et al., 2019; Liu et al., 2023). Therefore,

inhibiting the formation of bacterial biofilms is vital in reducing the incidence of foodborne disease. Many biomolecules have been shown to disrupt biofilms; however, the dynamic interactions that occur during this process have been poorly investigated. GROMACS provides an ideal computer tool for studying the interaction between biomolecules and biofilms, which can be used to characterize and investigate the dynamic changes that occur during this process and the detailed mechanism of the interaction between biomolecules and biofilms. Neto and Cordeiro (2016) investigated the impact of phospholipid peroxidation and cholesterol peroxidation on the properties of 1-palmitoyl-2-oleoyl-1-palmitoyl-*sn*-glycero-3-phosphocholine (POPC) lipid bilayers using MD simulation. The results demonstrated that peroxidation of phospholipids and cholesterol through either individual or combined peroxidation led to migration of the —OOH groups formed in the lipids to the membrane surface. This migration causes the *sn*-2 acyl chains of phospholipids to bend, while in the case of cholesterol hydroperoxides, the sterol backbone is tilted due to its rigidity. Lyu et al. (2018) explored the interaction mechanism of curcumin (CUR) on various types of model lipid bilayers and biological membrane mimics. The results indicated that CUR enters these lipid bilayer systems, particularly at locations close to the lipid tails and at the interface between the lipid head and tail. It was also observed that the orientation of CUR was mostly parallel to the membrane surface. This simulation revealed that CUR has different destructive effects on various microbial membranes. Alghamdi et al. (2019) elucidated the antifungal mechanism of mung bean defense factor VrD1 through GROMACS simulation. When VrD1 interacts with the di-phosphatidyl-choline (DPPC) lipid bilayer, the ordered structure of the DPPC lipid bilayer is disrupted, and this interferes with normal cellular physiological processes and ultimately leads to cell death. Nie et al. (2021) compared the bactericidal activity of chlorogenic acid and isochlorogenic acid A by simulating their interactions with POPC and 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphoglycerol (POPG) lipid bilayers. As demonstrated in Fig. 4, isochlorogenic acid A inserted more deeply into the lipid bilayers than did chlorogenic acid, which explained why isochlorogenic acid A has stronger bactericidal activity than does chlorogenic acid at the molecular level.

Overall, lipids have a variety of forms and applications in food. Through GROMACS molecular dynamics simulation, researchers can understand the dynamic processes of lipid nanostructures from the perspective of molecules and atoms. GROMACS can also characterize the formation and properties of emulsion gels used to enhance the taste and acceptability of food. Moreover, GROMACS simulation can be used to explore the mechanisms of interaction between biomolecules and biomembranes, which is vital for improving food safety and quality.

3.4. Small molecules

The numerous bioactive small molecules present in food, such as flavonoids, anthocyanins and carotenoids, possess many functional properties. However, their applications in the food industry are limited due to their poor solubility, weak stability and low bioavailability. A previous study (McClements et al., 2015) demonstrated that incorporating these bioactive compounds into suitable delivery systems is an effective strategy for enhancing the stability and solubility of these substances. Proteins contain various functional groups and high-affinity hydrophobic binding sites due to their unique molecular structures, making them suitable for delivering ligands and bioactive compounds (Teimouri et al., 2022). Hence, GROMACS molecular dynamics simulation provides valuable guidance for understanding the embedding mechanisms of the bioactive compounds and for screening the optimal protein carriers of bioactive compounds. It has been suggested that β-lactoglobulin might be a natural carrier for various bioactive compounds. Gholami and Bordbar (2014) explored the binding characteristics of naringenin and β-lactoglobulin by GROMACS simulation. A stable complex between naringenin and β-lactoglobulin was observed, indicating that the two compounds possess significant binding affinity. Aprodu et al. (2017) found that the binding site of sea buckthorn carotenoids changed under high temperature conditions, which caused sea buckthorn carotenoids and β-lactoglobulin to form a more stable complex through hydrophobic interactions. Wang et al. (2021) investigated the pH-dependent interaction between lycopene and β-lactoglobulin and explored how pH affects their binding ability. The results indicated that

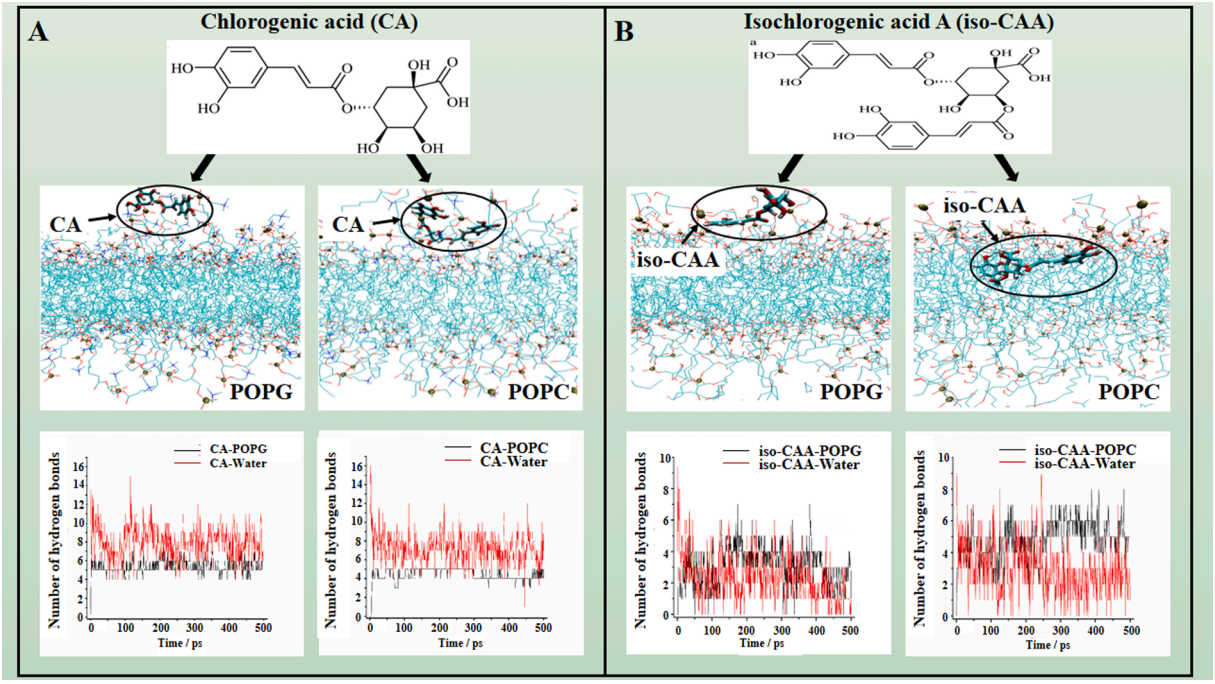


Fig. 4. The interaction mechanism between the two polyphenols and the two model lipid bilayers (Nie et al., 2021). (A) The binding configurations, locations and number of hydrogen bonds of CA on POPG and POPC lipid bilayers. (B) The binding configurations, locations and number of hydrogen bonds of *iso*-CAA on POPG and POPC lipid bilayers.

at pH 8.1, the binding affinity and the structural stability of the complex between lycopene and β-lactoglobulin improved compared to that at pH 7.0. Pan et al. (2021) evaluated the binding modes and affinities of the complexes formed between C3G and four cow's milk (CM) proteins, including α-lactalbumin (α-LA), β-lactoglobulin (β-LG), α_{s1}-casein (α_{s1}-CA) and β-casein (β-CA). As shown in Fig. 5, the molecular dynamics simulation results for RMSD, Rg, SASA and number of hydrogen bonds indicated that, among the four proteins of complexes, the C3G-β-CA complex exhibited the most compact and stable structure, proving that β-CA was the most promising protein carrier for improving the stability and bioavailability of C3G.

In the food industry, food additives are used to improve sensory properties of food, enhance its flavor and extend its shelf life (Wu et al., 2022). However, once ingested, they are digested and absorbed through the gastrointestinal tract, where they interact with biological molecules in the human body. Therefore, investigation of the interactions between food additives and biomolecules such as proteins is extremely valuable for revealing the potential adverse effects of food additives on the human body. GROMACS simulation technology can be used to visualize the three-dimensional conformations of complexes between food additives and biological molecules, to explore their binding mechanisms, and to identify key binding sites at the atomic scale. Mohseni-Shahri et al. (2018) studied the effect of sunset yellow (SY) on the activity of pepsin. They found that the predominant site at which SY binds to pepsin is the tyrosine residue located at the entrance to the pepsin active site and that the SY-pepsin complex was held together through stacking interactions and hydrogen bonding. Moreover, the simulation results also showed that SY could inhibit the activity of pepsin by preventing substrate entry. Jafari-Arvari et al. (2021) conducted a detailed investigation into the interaction between the food colorant tartrazine and apo-transferrin using GROMACS molecular dynamics simulation. This study revealed significant binding between tartrazine and apo-transferrin and showed that the binding led to changes in protein conformation. Hashemi-Shahraki et al. (2021) found that the secondary and tertiary structures of pepsin appear to be greatly affected by the formation of a complex between the protein and quinoline yellow and that formation of the complex was driven by hydrogen bonds and van der Waals interactions. Sadat Mostafavi et al. (2022) analyzed the binding between Ponceau 4R and trypsin through GROMACS simulation. They found that the stability and compactness of trypsin were reduced after combination with Ponceau 4R.

GROMACS simulation technology has been proven to possess many merits that make it helpful in comprehending the mechanisms of embedding of bioactive compounds and selecting the optimal protein carriers for these substances. This technique is also useful for predicting interactions and identifying the key binding sites between food additives and biological macromolecules and thus evaluating the safety of food additives and elucidating their potential effects on human health.

4. Conclusion and prospects

GROMACS molecular dynamics simulation has proven to be an invaluable tool in food science research. This review provides a comprehensive understanding of the applications of GROMACS molecular dynamics simulations of food components, including proteins, carbohydrates, lipids and small molecular compounds, and shows that it can provide profound insights into the intricate behaviors of and interactions between these biomolecules. GROMACS molecular dynamics simulation can be used to observe the dynamic processes undergone by food ingredients and to reveal the interaction mechanisms involved and the structural stabilities of various components. Furthermore, the GROMACS molecular dynamics simulation technique is able to deeply explore the behavior of food molecules under different processing and external conditions, such as the effects of temperature, pressure and pH value on food structure, which is highly important in determining food processing and storage conditions.

Although the GROMACS simulation method plays an important role in the food science research, there are still some significant challenges and limitations. Primarily, GROMACS simulation should possess greater computational capacity; it requires a large amount of computing resources due to the highly complicated molecular structure of food. Moreover, the accuracy of the force field is the main factor that affects

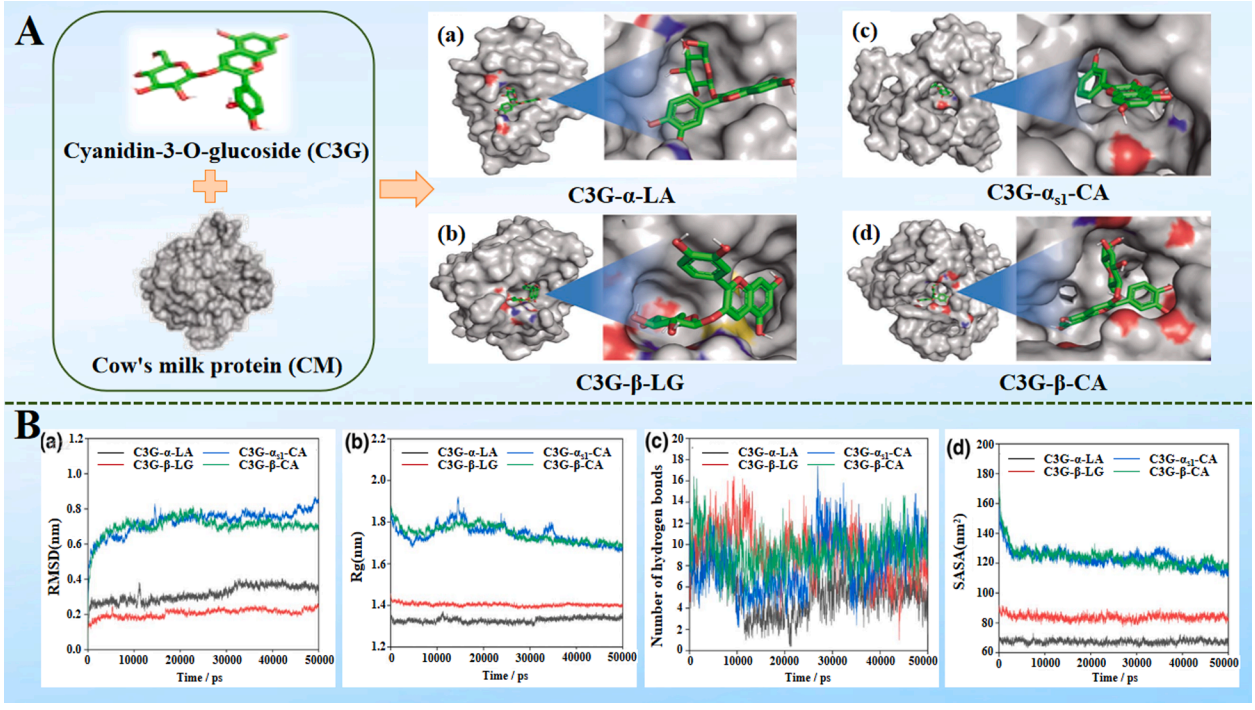


Fig. 5. The interaction of cyanidin-3-O-glucoside (C3G) and four cow's milk proteins (CM) (Pan et al., 2021). (A) The binding conformation of C3G and four CM protein complexes. (a) C3G-α-lactalbumin; (b) C3G-β-lactoglobulin; (c) C3G-α_{s1}-casein; (d) C3G-β-casein. (B) The molecular dynamics simulation results of C3G-CM complexes. (a) Root mean square deviations (RMSD); (b) Radius of gyration (Rg); (c) Number of hydrogen bonds; (d) Solvent accessible surface area (SASA).

the reliability of GROMACS molecular dynamics simulation results; thus, the specific force fields are required for some food molecules. For example, the force fields of some carbohydrates, such as polysaccharide molecules, often needs to be transformed and processed through other methods. Consequently, more attention should be given to optimizing the force fields used in GROMACS molecular dynamics simulation and its computational performance, which will increase its applicability to food science research. In the future, we hope that researchers will continue to explore and expand the application of GROMACS molecular dynamics simulations in food science research. It is also necessary to combine the GROMACS method with traditional experimental methods used in scientific research; this not only provides more accurate guidance and more comprehensive insights but also significantly propels the development of food science research.

CRediT authorship contribution statement

Dongping Yu: Writing – review & editing, Writing – original draft, Conceptualization. **Haiping Li:** Supervision, Resources. **Yuzi Liu:** Writing – review & editing. **Xingqun Yang:** Methodology, Investigation. **Wei Yang:** Investigation. **Yiran Fu:** Investigation. **Yi-ao Zuo:** Investigation. **Xianya Huang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported by the Tianjin Natural Science Foundation (No. 6232009) and Tianjin University of Commerce College Student Innovation and Entrepreneurship Training Program (202310069228; 202410069007).

References

Abdollahi, K., Condict, L., Hung, A., & Kasapis, S. (2022). Binding parameters and molecular dynamics of β-lactoglobulin-vanillic acid complexation as a function of pH - part B: Neutral pH. *Food Chemistry, 367*, Article 130655. https://doi.org/10.1016/j.foodchem.2021.130655

Abdollahi, K., Condict, L., Hung, A., & Kasapis, S. (2023). Examination of β-lactoglobulin-ferulic acid complexation at elevated temperature using biochemical spectroscopy, proteomics and molecular dynamics. *Food Hydrocolloids, 134*, Article 108053. https://doi.org/10.1016/j.foodhyd.2022.108053

Abraham, M. J., Murtola, T., Schulz, R., Páll, S., Smith, J. C., Hess, B., & Lindahl, E. (2015). GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX, 1–2*, 19–25. https://doi.org/10.1016/j.softx.2015.06.001

Alghamdi, H. A., Campbell, L. J., & Euston, S. R. (2019). Molecular dynamics simulation of the adsorption of mung bean defensin VrD1 to a phospholipid bilayer. *Food Structure, 21*, Article 100117. https://doi.org/10.1016/j.foostr.2019.100117

Alvarez-Ordonez, A., Coughlan, L. M., Briandet, R., & Cotter, P. D. (2019). Biofilms in food processing environments: Challenges and opportunities. *Annual Review of Food Science and Technology, 10*, 173–195. https://doi.org/10.1146/annurev-food-032818-121805

Aprodu, I., Ursache, F., Turturică, M., Râpeanu, G., & Stănciuc, N. (2017). Thermal stability of the complex formed between carotenoids from sea buckthorn (*Hippophae rhamnoides L.*) and bovine β-lactoglobulin. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, 173*, 562–571. https://doi.org/10.1016/j.saa.2016.10.010

Batista, M. L. S., Pérez-Sánchez, G., Gomes, J. R. B., Coutinho, J. A. P., & Maginn, E. J. (2015). Evaluation of the GROMOS $56A_{CARBO}$ force field for the calculation of structural, volumetric, and dynamic properties of aqueous glucose systems. *The Journal of Physical Chemistry B, 119*(49), 15310–15319. https://doi.org/10.1021/acs.jpcb.5b08155

Bhopatkar, D., Feng, T., Chen, F., Zhang, G., Carignano, M., Park, S. H., Zhuang, H., Campanella, O. H., & Hamaker, B. R. (2015). Self-assembled nanoparticle of common food constituents that carries a sparingly soluble small molecule. *Journal of Agricultural and Food Chemistry, 63*(17), 4312–4319. https://doi.org/10.1021/acs.jafc.5b00037

Cen, C., Chen, J., Wang, W., Zhang, J., Yang, X., Fu, L., & Wang, Y. (2022). Exploring the interaction mechanism of dietary protein ovalbumin and folic acid: A combination research of molecular simulation technology and multispectroscopy. *Food Chemistry, 385*, Article 132536. https://doi.org/10.1016/j.foodchem.2022.132536

Chamizo-González, F., Estévez, I. G., Gordillo, B., Manjón, E., Escribano-Bailón, M. T., Heredia, F. J., & González-Miret, M. L. (2023). First insights into the binding mechanism and colour effect of the interaction of grape seed 11S globulin with malvidin 3-O-glucoside by fluorescence spectroscopy, differential colorimetry and molecular modelling. *Food Chemistry, 413*, Article 135591. https://doi.org/10.1016/j.foodchem.2023.135591

Chen, G., Huang, K., Miao, M., Feng, B., & Campanella, O. H. (2019). Molecular dynamics simulation for mechanism elucidation of food processing and safety: State of the art. *Comprehensive Reviews in Food Science and Food Safety, 18*(1), 243–263. https://doi.org/10.1111/1541-4337.12406

Chen, G., Miao, M., Jiang, B., Jin, J., Campanella, O. H., & Feng, B. (2017). Effects of high hydrostatic pressure on lipase from Rhizopus chinensis: I. Conformational changes. *Innovative Food Science & Emerging Technologies, 41*, 267–276. https://doi.org/10.1016/j.ifset.2017.03.016

Chen, Y., Nie, H., Dong, H., Gao, Q., Peng, X., Wang, N., Chen, X., Liu, Q., Li, J., Xu, X., & Xue, Y. (2023). Revealing the mechanism underlying the effects of γ-aminobutyric acid-dioscorin interactions on dioscorin structure and emulsifying properties by molecular dynamic simulations. *Food Research International, 171*, Article 112982. https://doi.org/10.1016/j.foodres.2023.112982

Cheng, L., Feng, T., Zhang, B., Zhu, X., Hamaker, B., Zhang, H., & Campanella, O. (2018). A molecular dynamics simulation study on the conformational stability of amylose-linoleic acid complex in water. *Carbohydrate Polymers, 196*, 56–65. https://doi.org/10.1016/j.carbpol.2018.04.102

Cheng, W., Li, C., Xiao, F., He, J., Liu, L., Niu, H., & Ma, J. (2024). Elucidating binding mechanisms of caffeic acid and resveratrol by beta-lactoglobulin: Insights into hydrophobic interactions and complex formation. *Food Hydrocolloids, 146*, Article 109269. https://doi.org/10.1016/j.foodhyd.2023.109269

Condict, L., Hung, A., Ashton, J., & Kasapis, S. (2021). High-temperature binding parameters and molecular dynamics of 4-hydroxybenzoic acid and β-casein complexes, determined via the method of continuous variation and fluorescence spectroscopy. *Food Hydrocolloids, 114*, Article 106567. https://doi.org/10.1016/j.foodhyd.2020.106567

Corredig, M., Young, N., & Dalsgaard, T. K. (2020). Food proteins: Processing solutions and challenges. *Current Opinion in Food Science, 35*, 49–53. https://doi.org/10.1016/j.cofs.2019.12.010

Da Silva Santos, V., Badan Ribeiro, A. P., & Andrade Santana, M. H. (2019). Solid lipid nanoparticles as carriers for lipophilic compounds for applications in foods. *Food Research International, 122*, 610–626. https://doi.org/10.1016/j.foodres.2019.01.032

Darmawan, K. K., Karagiannis, T. C., Hughes, J. G., Small, D. M., & Hung, A. (2020). High temperature induced structural changes of apo-lactoferrin and interactions with β-lactoglobulin and α-lactalbumin for potential encapsulation strategies. *Food Hydrocolloids, 105*, Article 105817. https://doi.org/10.1016/j.foodhyd.2020.105817

Darmawan, K. K., Karagiannis, T. C., Hughes, J. G., Small, D. M., & Hung, A. (2021). Effects of low temperatures on the conformation of apo-lactoferrin and its interactions with α-lactalbumin and β-lactoglobulin: Application of in silico approaches. *Food Hydrocolloids, 121*, Article 107055. https://doi.org/10.1016/j.foodhyd.2021.107055

Dasgupta, N., Ranjan, S., & Gandhi, M. (2019). Nanoemulsions in food: Market demand. *Environmental Chemistry Letters, 17*(2), 1003–1009. https://doi.org/10.1007/s10311-019-00856-2

Deng, C., Cao, C., Zhang, Y., Hu, J., Gong, Y., Zheng, M., & Zhou, Y. (2022). Formation and stabilization mechanism of β-cyclodextrin inclusion complex with C10 aroma molecules. *Food Hydrocolloids, 123*, Article 107013. https://doi.org/10.1016/j.foodhyd.2021.107013

Di, C., & Jia, W. (2023). Capture modes evaluation of four flavor constituents in β-lactoglobulin by high-resolution mass spectrometry and molecular dynamics simulation approaches. *Food Hydrocolloids, 145*, Article 109157. https://doi.org/10.1016/j.foodhyd.2023.109157

Dondapati, R. S. (2019). Overview of BIOVIA Materials Studio, LAMMPS, and GROMACS. In molecular dynamics simulation of nanocomposites using BIOVIA Materials Studio, Lammps and Gromacs. *The Netherlands*, 39–100. https://doi.org/10.1016/B978-0-12-816954-4.00002-4

Euston, S. R. (2013). Molecular dynamics simulation of the effect of heat on the conformation of bovine β-lactoglobulin A: A comparison of conventional and accelerated methods. *Food Hydrocolloids, 30*(2), 519–530. https://doi.org/10.1016/j.foodhyd.2012.07.016

Feng, T., Li, M., Zhou, J., Zhuang, H., Chen, F., Ye, R., Campanella, O., & Fang, Z. (2015). Application of molecular dynamics simulation in food carbohydrate research-a review. *Innovative Food Science & Emerging Technologies, 31*, 1–13. https://doi.org/10.1016/j.ifset.2015.06.015

Feng, X., Li, F., Ding, M., Zhang, R., & Shi, T. (2020). Molecular dynamic simulation: Conformational properties of single-stranded curdlan in aqueous solution. *Carbohydrate Polymers, 250*, Article 116906. https://doi.org/10.1016/j.carbpol.2020.116906

Filipe, H. A. L., & Loura, L. M. S. (2022). Molecular dynamics simulations: Advances and applications. *Molecules, 27*(7), 2105. https://doi.org/10.3390/molecules27072105

Geng, S., Jiang, Z., Ma, H., Pu, P., Liu, B., & Liang, G. (2021). Fabrication and characterization of novel edible Pickering emulsion gels stabilized by dihydromyricetin. *Food Chemistry, 343*, Article 128486. https://doi.org/10.1016/j.foodchem.2020.128486

Gholami, S., & Bordbar, A. (2014). Exploring binding properties of naringenin with bovine β-lactoglobulin: A fluorescence, molecular docking and molecular dynamics simulation study. *Biophysical Chemistry, 187–188*, 33–42. https://doi.org/10.1016/j.bpc.2014.01.003

González, M. A. (2011). Force fields and molecular dynamics simulations. *Collection Sfn, 12*, 169–200. https://doi.org/10.1051/sfn/201112009

Gui, H., Jiang, Q., Tian, J., Zang, Z., Yang, S., Yang, Y., Xin, M., Zhao, M., Dai, J., & Li, B. (2023). Interaction and binding mechanism of cyanidin-3-O-glucoside to lysozyme in varying pH conditions: Multi-spectroscopic, molecular docking and molecular dynamics simulation approaches. *Food Chemistry, 425*, Article 136509. https://doi.org/10.1016/j.foodchem.2023.136509

Guo, S., Ma, C., Hu, Y., Bai, G., Song, Z., & Cao, X. (2022). Solid lipid nanoparticles for phytosterols delivery: The acyl chain number of the glyceride matrix affects the arrangement, stability, and release. *Food Chemistry, 394*, Article 133412. https://doi.org/10.1016/j.foodchem.2022.133412

Han, P., An, N., Yang, L., Ren, X., Lu, S., Ji, H., Wang, Q., & Dong, J. (2022). Molecular dynamics simulation of the interactions between sesamol and myosin combined with spectroscopy and molecular docking studies. *Food Hydrocolloids, 131*, Article 107801. https://doi.org/10.1016/j.foodhyd.2022.107801

Hashemi-Shahraki, F., Shareghi, B., & Farhadian, S. (2021). Characterizing the binding affinity and molecular interplay between quinoline yellow and pepsin. *Journal of Molecular Liquids, 341*, Article 117317. https://doi.org/10.1016/j.molliq.2021.117317

Hess, B., Kutzner, C., Van der Spoel, D., & Lindahl, E. (2008). GROMACS 4: Algorithms for highly efficient, load-balanced, and scalable molecular simulation. *Journal of Chemical Theory and Computation, 4*(3), 435–447. https://doi.org/10.1021/ct700301q

Hollingsworth, S. A., & Dror, R. O. (2018). Molecular dynamics simulation for all. *Neuron, 99*(6), 1129–1143. https://doi.org/10.1016/j.neuron.2018.08.011

Hu, X., Zeng, Z., Zhang, J., Wu, D., Li, H., & Geng, F. (2023). Molecular dynamics simulation of the interaction of food proteins with small molecules. *Food Chemistry, 405*, Article 134824. https://doi.org/10.1016/j.foodchem.2022.134824

Huang, W., Dalal, I. S., & Larson, R. G. (2014). Analysis of solvation and gelation behavior of methylcellulose using atomistic molecular dynamics simulations. *The Journal of Physical Chemistry B, 118*(48), 13992–14008. https://doi.org/10.1021/jp509760x

Huang, Z., Ni, D., Chen, Z., Zhu, Y., Zhang, W., & Mu, W. (2023). Application of molecular dynamics simulation in the field of food enzymes: Improving the thermal-stability and catalytic ability. *Critical Reviews in Food Science and Nutrition*, 1–13. https://doi.org/10.1080/10408398.2023.2238054

Jafari-Arvari, H., Saei-Dehkordi, S. S., & Farhadian, S. (2021). Evaluation of interactions between food colorant, tartrazine, and Apo-transferrin using spectroscopic analysis and docking simulation. *Journal of Molecular Liquids, 339*, Article 116715. https://doi.org/10.1016/j.molliq.2021.116715

Jiang, Z., Lu, X., Geng, S., Ma, H., & Liu, B. (2020). Structuring of sunflower oil by stearic acid derivatives: Experimental and molecular modelling studies. *Food Chemistry, 324*, Article 126801. https://doi.org/10.1016/j.foodchem.2020.126801

K., S., & Kumar, A. (2022). Nanoemulsions: Techniques for the preparation and the recent advances in their food applications. *Innovative Food Science & Emerging Technologies, 76*, 102914. https://doi.org/10.1016/j.ifset.2021.102914

Kumar, P., Bhardwaj, V. K., & Purohit, R. (2023). Dispersion-corrected DFT calculations and umbrella sampling simulations to investigate stability of chrysin-cyclodextrin inclusion complexes. *Carbohydrate Polymers, 319*, Article 121162. https://doi.org/10.1016/j.carbpol.2023.121162

Li, Y., Yuan, Z., Gao, Y., Bao, Z., Sun, N., & Lin, S. (2022). Mechanism of trypsin activation by pulsed electric field treatment revealed based on chemical experiments and molecular dynamics simulations. *Food Chemistry, 394*, Article 133477. https://doi.org/10.1016/j.foodchem.2022.133477

Liang, F., Shi, Y., Shi, J., & Cao, W. (2023). Exploring the binding mechanism of pumpkin seed protein and apigenin: Spectroscopic analysis, molecular docking and molecular dynamics simulation. *Food Hydrocolloids, 137*, Article 108318. https://doi.org/10.1016/j.foodhyd.2022.108318

Liu, X., Yao, H., Zhao, X., & Ge, C. (2023). Biofilm formation and control of foodborne pathogenic bacteria. *Molecules, 28*(6), 2432. https://doi.org/10.3390/molecules28062432

Liu, Z., Fu, Y., Zhang, F., Zhao, Q., Xue, Y., Hu, J., & Shen, Q. (2022). Comparison of the molecular structure of heat and pressure-treated corn starch based on experimental data and molecular dynamics simulation. *Food Hydrocolloids, 125*, Article 107371. https://doi.org/10.1016/j.foodhyd.2021.107371

Liu, Z., Junejo, S. A., Zhang, B., Fu, X., & Huang, Q. (2022). Characteristics and ethylene encapsulation properties of V-type linear dextrin with different degrees of polymerisation. *Carbohydrate Polymers, 277*, Article 118814. https://doi.org/10.1016/j.carbpol.2021.118814

Lyu, Y., Xiang, N., Mondal, J., Zhu, X., & Narsimhan, G. (2018). Characterization of interactions between curcumin and different types of lipid bilayers by molecular dynamics simulation. *The Journal of Physical Chemistry B, 122*(8), 2341–2354. https://doi.org/10.1021/acs.jpcb.7b10566

Mackerell, A. D. (2004). Empirical force fields for biological macromolecules: Overview and issues. *Journal of Computational Chemistry, 25*(13), 1584–1604. https://doi.org/10.1002/jcc.20082

McClements, D. J., Zou, L., Zhang, R., Salvia-Trujillo, L., Kumosani, T., & Xiao, H. (2015). Enhancing nutraceutical performance using excipient foods: Designing food structures and compositions to increase bioavailability. *Comprehensive Reviews in Food Science and Food Safety, 14*(6), 824–847. https://doi.org/10.1111/1541-4337.12170

Meng, Y., Wei, Z., & Xue, C. (2023). Deciphering the interaction mechanism and binding mode between chickpea protein isolate and flavonoids based on experimental studies and molecular simulation. *Food Chemistry, 429*, Article 136848. https://doi.org/10.1016/j.foodchem.2023.136848

Moghaddasi, F., Housaindokht, M. R., Darroudi, M., Bozorgmehr, M. R., & Sadeghi, A. (2018). Soybean oil-based nanoemulsion systems in absence and presence of curcumin: Molecular dynamics simulation approach. *Journal of Molecular Liquids, 264*, 242–252. https://doi.org/10.1016/j.molliq.2018.05.066

Mohseni-Shahri, F. S., Moeinpour, F., & Nosrati, M. (2018). Spectroscopy and molecular dynamics simulation study on the interaction of sunset yellow food additive with pepsin. *International Journal of Biological Macromolecules, 115*, 273–280. https://doi.org/10.1016/j.ijbiomac.2018.04.080

Neto, A. J. P., & Cordeiro, R. M. (2016). Molecular simulations of the effects of phospholipid and cholesterol peroxidation on lipid membrane properties. *Biochimica et Biophysica Acta (Bba) – Biomembranes, 1858*(9), 2191–2198. https://doi.org/10.1016/j.bbamem.2016.06.018

Nian, B., Xu, Y., & Liu, Y. (2021). Molecular dynamics simulation for mechanism revelation of the safety and nutrition of lipids and derivatives in food: State of the art. *Food Research International, 145*, Article 110399. https://doi.org/10.1016/j.foodres.2021.110399

Nie, H., Dong, H., Chen, Y., Hao, M., Chen, J., Tang, Z., Liu, Q., Li, J., Xu, X., & Xue, Y. (2023). Effects of spray drying and freeze drying on the structure and emulsifying properties of yam soluble protein: A study by experiment and molecular dynamics simulation. *Food Chemistry, 409*, Article 135238. https://doi.org/10.1016/j.foodchem.2022.135238

Nie, R., Dang, M., Ge, Z., Huo, Y., Yu, B., & Tang, S. (2021). Interactions of chlorogenic acid and isochlorogenic acid A with model lipid bilayer membranes: Insights from molecular dynamics simulations. *Chemistry and Physics of Lipids, 240*, Article 105136. https://doi.org/10.1016/j.chemphyslip.2021.105136

Nnyigide, O. S., & Hyun, K. (2020). The protection of bovine serum albumin against thermal denaturation and gelation by sodium dodecyl sulfate studied by rheology and molecular dynamics simulation. *Food Hydrocolloids, 103*, Article 105656. https://doi.org/10.1016/j.foodhyd.2020.105656

Pall, S., Zhmurov, A., Bauer, P., Abraham, M., Lundborg, M., Gray, A., Hess, B., & Lindahl, E. (2020). Heterogeneous parallelization and acceleration of molecular dynamics simulations in GROMACS. *Journal of Chemical Physics, 153*(13), Article 134110. https://doi.org/10.1063/5.0018516

Pan, F., Li, J., Zhao, L., Tuersuntuoheti, T., Mehmood, A., Zhou, N., Hao, S., Wang, C., Guo, Y., & Lin, W. (2021). A molecular docking and molecular dynamics simulation study on the interaction between cyanidin-3-O-glucoside and major proteins in cow's milk. *Journal of Food Biochemistry, 45*(1), Article e13570. https://doi.org/10.1111/jfbc.13570

Peng, Y., Wang, C., Yu, J., Wu, J., Wang, F., Liu, Y., & Li, X. (2023). Self-assembly mechanism of rice glutelin amyloid fibril aggregates obtained through experimental and molecular dynamics simulation analysis. *Food Hydrocolloids, 143*, Article 108867. https://doi.org/10.1016/j.foodhyd.2023.108867

Premjit, Y., Pandhi, S., Kumar, A., Rai, D. C., Duary, R. K., & Mahato, D. K. (2022). Current trends in flavor encapsulation: A comprehensive review of emerging encapsulation techniques, flavour release, and mathematical modelling. *Food Research International, 151*, Article 110879. https://doi.org/10.1016/j.foodres.2021.110879

Sadat Mostafavi, E., Asoodeh, A., & Chamani, J. (2022). Evaluation of interaction between Ponceau 4R (P4R) and trypsin using kinetic, spectroscopic, and molecular dynamics simulation methods. *Journal of Molecular Liquids, 362*, Article 119761. https://doi.org/10.1016/j.molliq.2022.119761

Shao, Z., Fang, S., Li, Y., Chen, J., & Meng, Y. (2018). Physicochemical properties and formation mechanism of electrostatic complexes based on ε-polylysine and whey protein: Experimental and molecular dynamics simulations study. *International Journal of Biological Macromolecules, 118*, 2208–2215. https://doi.org/10.1016/j.ijbiomac.2018.07.086

Singh, A., Vanga, S. K., Orsat, V., & Raghavan, V. (2018). Application of molecular dynamic simulation to study food proteins: A review. *Critical Reviews in Food Science and Nutrition, 58*(16), 2779–2789. https://doi.org/10.1080/10408398.2017.1341864

Smith, A., Dong, X., & Raghavan, V. (2022). An overview of molecular dynamics simulation for food products and processes. *Processes, 10*(1), 119. https://doi.org/10.3390/pr10010119

Stănciuc, N., Banu, I., Bolea, C., Patraşcu, L., & Aprodu, I. (2018). Structural and antigenic properties of thermally treated gluten proteins. *Food Chemistry, 267*, 43–51. https://doi.org/10.1016/j.foodchem.2017.03.018

Stănciuc, N., Creţu, A. A., Banu, I., & Aprodu, I. (2018). Advances on the impact of thermal processing on structure and antigenicity of chicken ovomucoid. *Journal of the Science of Food and Agriculture, 98*(8), 3119–3128. https://doi.org/10.1002/jsfa.8813

Teimouri, S., Kasapis, S., & Dokouhaki, M. (2022). Diffusional characteristics of food protein-based materials as nutraceutical delivery systems: A review. *Trends in Food Science & Technology, 122*, 201–210. https://doi.org/10.1016/j.tifs.2022.02.025

Tian, J., Zhao, M., Xu, J., Lv, T., Liu, X., Sun, S., Guan, Q., Zhou, Z., Wu, J., Zhao, M., Li, Y., Liu, H., Niu, S., & Hu, P. (2023). Inhibitory mechanism of prenylated flavonoids isolated from mulberry leaves on α-glucosidase by multi-spectroscopy and molecular dynamics simulation. *Journal of Agricultural and Food Chemistry, 71*(23), 9135–9147. https://doi.org/10.1021/acs.jafc.3c00776

Van der Spoel, D., Lindahl, E., Hess, B., Groenhof, G., Mark, A. E., & Berendsen, H. J. (2005). GROMACS: Fast, flexible, and free. *Journal of Computational Chemistry, 26* (16), 1701–1718. https://doi.org/10.1002/jcc.20291

Wang, C., Chen, L., Lu, Y., Liu, J., Zhao, R., Sun, Y., Sun, B., & Cuina, W. (2021). pH-Dependent complexation between β-lactoglobulin and lycopene: Multi-spectroscopy, molecular docking and dynamic simulation study. *Food Chemistry, 362*, Article 130230. https://doi.org/10.1016/j.foodchem.2021.130230

Wang, H., Zhu, J., Zhang, H., Chen, Q., & Kong, B. (2022). Understanding interactions among aldehyde compounds and porcine myofibrillar proteins by spectroscopy and molecular dynamics simulations. *Journal of Molecular Liquids, 349*, Article 118190. https://doi.org/10.1016/j.molliq.2021.118190

Wang, J., Yu, Z., Wu, W., He, S., Xie, B., Wu, M., & Sun, Z. (2022). Molecular mechanism of epicatechin gallate binding with carboxymethyl β-glucan and its effect on antibacterial activity. *Carbohydrate Polymers, 298*, Article 120105. https://doi.org/10.1016/j.carbpol.2022.120105

Wang, R., Fan, R., Meng, T., & Wang, L. (2023). Exploration of the inhibitory mechanisms of trans-polydatin/resveratrol on α-glucosidase by multi-spectroscopic analysis, *in silico* docking and molecular dynamics simulation. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, 299*, Article 122866. https://doi.org/10.1016/j.saa.2023.122866

Wang, X., Deng, Y., Zhang, Y., Zhang, C., Liu, L., Liu, Y., Jiang, J., Xie, P., & Huang, L. (2023). Screening and evaluation of novel α-glucosidase inhibitory peptides from *ginkgo biloba* seed cake based on molecular docking combined with molecular dynamics simulation. *Journal of Agricultural and Food Chemistry, 71*(27), 10326–10337. https://doi.org/10.1021/acs.jafc.3c00826

Westberry, B. P., Mansel, B. W., Ryan, T. M., Lundin, L., & Williams, M. A. K. (2022). X-ray scattering and molecular dynamics simulations reveal the secondary structure of κ-carrageenan in the solution state. *Carbohydrate Polymers, 296*, Article 119958. https://doi.org/10.1016/j.carbpol.2022.119958

Wu, L., Zhang, C., Long, Y., Chen, Q., Zhang, W., & Liu, G. (2022). Food additives: From functions to analytical methods. *Critical Reviews in Food Science and Nutrition, 62*(30), 8497–8517. https://doi.org/10.1080/10408398.2021.1929823

Wu, Q., Zhang, C., Dong, W., Lu, H., Yang, Y., Li, W., Xu, Y., & Li, X. (2023). Simultaneously enhanced thermostability and catalytic activity of xylanase from *streptomyces rameus* L2001 by rigidifying flexible regions in loop regions of the N-terminus. *Journal of Agricultural and Food Chemistry, 71*(34), 12785–12796. https://doi.org/10.1021/acs.jafc.3c03871

Xu, H., Zhang, S., & Yu, W. (2022). Revealing the mechanism beneath the effects of starch-amino acids interactions on starch physicochemical properties by molecular dynamic simulations. *Food Hydrocolloids, 124*, Article 107359. https://doi.org/10.1016/j.foodhyd.2021.107359

Yan, B., Chen, T., Tao, Y., Zhang, N., Zhao, J., Zhang, H., Chen, W., & Fan, D. (2023). Conformation and hydration property of low-acetyl gellan gum under microwave irradiation: Experiments and molecular dynamics simulations. *Food Hydrocolloids, 145*, Article 109140. https://doi.org/10.1016/j.foodhyd.2023.109140

Yang, P., Wang, W., Xu, Z., Rao, L., Zhao, L., Wang, Y., et al. (2023). New insights into the pH dependence of anthocyanin-protein interactions by a case study of cyanidin-3-O-glucoside and bovine serum albumin. *Food Hydrocolloids, 140*, Article 108649. https://doi.org/10.1016/j.foodhyd.2023.108649

Ye, Y., Zhang, H., Deng, J., Li, M., & Chen, Z. (2022). L-Arginine inhibits the activity of α-amylase: Rapid kinetics, interaction and functional implications. *Food Chemistry, 380*, Article 131836. https://doi.org/10.1016/j.foodchem.2021.131836

Yu, Y., Xu, S., He, R., & Liang, G. (2023). Application of molecular simulation methods in food science: Status and prospects. *Journal of Agricultural and Food Chemistry, 71*(6), 2684–2703. https://doi.org/10.1021/acs.jafc.2c06789

Zhang, B., Zhang, X., Shen, C., & Deng, S. (2018). Understanding the influence of carrageenan oligosaccharides and xylooligosaccharides on ice-crystal growth in peeled shrimp (Litopenaeus vannamei) during frozen storage. *Food & Function, 9*(8), 4394–4403. https://doi.org/10.1039/C8FO00364E

Zhou, H., Wang, F., Niu, H., Yuan, L., Tian, J., Cai, S., Bi, X., & Zhou, L. (2022). Structural studies and molecular dynamic simulations of polyphenol oxidase treated by high pressure processing. *Food Chemistry, 372*, Article 131243. https://doi.org/10.1016/j.foodchem.2021.131243

Zhu, J., Zhang, B., Tan, C., Ding, L., Shao, M., Chen, C., Fu, X., & Huang, Q. (2022). Effect of *Rosa Roxburghii* juice on starch digestibility: A focus on the binding of polyphenols to amylose and porcine pancreatic α-amylase by molecular modeling. *Food Hydrocolloids, 123*, Article 106966. https://doi.org/10.1016/j.foodhyd.2021.106966